

Julian Henry

Houston, TX USA · github.com/juleshenry · juleshenry.github.io

Full-stack software engineer building reliable, data-intensive systems across financial services, agriculture, cybersecurity, and applied AI.

Work Experience

Senior Full-Stack Engineer

October 2024 – Present

Macquarie Financial Ltd. — Houston, TX · Full-time

- Operate Java servlet services for critical trading workflows across London, São Paulo, and Sydney.
- Port as-dealt C++ deal logic to Python for ASIC reporting and auditable trade state.
- Co-author shared risk and workflow controls across 100+ commodity and financial products.
- Ship C++17 features in a 1M+ line platform for low-latency, high-availability trading.

Software Engineer

May 2023 – October 2024

AGCO Corporation — Atlanta, GA · Full-time

- Hardened AWS/Kafka telemetry for connected farm equipment across 140 countries.
- Engineered RESTful APIs for hardware device orchestration, accelerating integration-test cycles across connected farm equipment.
- Shipped Angular features for the Fendt.One precision agriculture platform supporting 10,000+ remote farm operations worldwide.

Data Engineer

August 2020 – August 2022

SimSpace Corporation — Boston, MA · Full-time

- Stress-tested ML threat-detection models through adversarial red-team simulations.
- Built cybersecurity analytics pipelines with NLP that improved threat detection rates by 72% across enterprise security workloads.
- Optimized a 30,000+ line Python/Pandas ETL pipeline, delivering 10× throughput.

Assets & Liabilities Intern

June 2019 – May 2020

Transamerica Corporation — Baltimore, MD · Internship → part-time

- Validated variable-annuity risk models with NumPy, scikit-learn, and pandas.
- Achieved 99.7% simulation accuracy for reconciliation and stress testing on a \$6B+ AUM variable annuity book.

Technical Skills

Languages: C++17, Python, Java, TypeScript, JavaScript, SQL, Julia, Rust, WebAssembly

Frontend: React, Next.js, Angular, HTML5, CSS3

Backend: Spring Boot, Node.js, Express, REST APIs, Java Servlets

Cloud / Infra: AWS, Kafka, Docker, Terraform, Linux, GitHub Actions, CI/CD, Cloudflare

Databases: PostgreSQL, MongoDB, DynamoDB, Neo4j, SurrealDB, SQLite, FAISS, Vector DBs

Data / ML: Pandas, NumPy, scikit-learn, TensorFlow, PySpark, Tableau

LLM Systems: RAG, LangGraph, structured outputs, tool calling

Education & Certificates

Bachelor of Applied Science (BAS) in Computer Science

University of Pennsylvania

- Vice President, Artificial Intelligence Society; [undergraduate thesis on differential privacy](#) under Rajeev Alur, with distinction
- **DELE B2** (Instituto Cervantes) — official upper-intermediate Spanish language credential
- **AWS Certified AI Practitioner** — Verification ID: TBD

Portfolio

- [One World Gazette \(1wg.ai\)](#) — a Next.js news platform using RAG for AI-powered global coverage.
- [glottosphere.com](#) — a Next.js dictionary using FAISS similarity search and a binary lexicon engine.
- [ghee](#) — a Python/Rich CLI with Bash and Zsh modules for 616+ shortcuts.
- [tatugem](#) — a Python CLI using Pillow and NumPy to generate recursive ASCII signatures.
- [readme-rosetta](#) — a Python CLI using Ollama, Rich, tqdm, and polib to translate Markdown READMEs.
- [intersekt](#) — WAT/WebAssembly big-integer arithmetic benchmarked against JavaScript BigInt.
- [quantum-plankton-ml](#) — TensorFlow Quantum and Cirq experiments compared with PyTorch CNN baselines.
- [vulgultra](#) — Python G2P/PanPhon candidate prep feeding a Rust simulated-annealing optimizer.